

## Batch Normalization Analysis - Problem 3

Feature means: [13. 5. 0.]

Feature variances: [ 1. 25. 25.]

Part (a):  $\hat{z}$  matrix (3x4):

```
[[-1.  1.  1. -1.]  
 [-1.  1.  1. -1.]  
 [-1.  1.  1. -1.]]
```

Part (b):  $\tilde{z}$  matrix (3x4):

```
[[ -1   1   1  -1]  
 [-11  -9  -9 -11]  
 [  9  11  11   9]]
```

Part (c): Differences between batch normalization during training and testing:

1. During training:

- Batch normalization uses statistics (mean and variance) computed from the current mini-batch

- These batch statistics are used for normalization in the forward pass

- Running averages of means and variances are updated for later use in testing

- This introduces noise that acts as regularization and helps with training

2. During testing:

- Instead of batch statistics, the stored population statistics (running mean and variance)

- accumulated during training are used

- This ensures consistent normalization regardless of test batch size

- No further updates to the running statistics occur

- The model behaves deterministically, without the regularizing noise of batch statistics

Part (d): How batch size during testing affects testing results:

1. When using population statistics (standard practice):

- The batch size during testing should have no effect on results

- Each sample is normalized using the same fixed statistics regardless of batch size

- This allows for consistent inference even with a single sample

2. If batch statistics were used during testing (not standard practice):

- Smaller batch sizes would lead to less reliable statistics

- Results would be inconsistent and depend on which samples are batched together

- Performance would likely degrade as batch size decreases

- Single-sample inference would be impossible as variance cannot be calculated

- Different inferences on the same input would be possible depending on the batch